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Editors
Infrared Physics & Technology

Dear Editors,

Please consider our manuscript, "Confidence-Shielded Evaluator-Conditioned Fusion for Risk-Controlled Infrared–Visible Object Detection," for publication in *Infrared Physics & Technology*.

Infrared–visible fusion can make thermal evidence available to object detectors, but the same fused image does not benefit every detector. We show that fusion utility changes across detector checkpoints and formulate selection as a label-free input-arbitration problem. The proposed Confidence-Shielded Evaluator-Conditioned Fusion (CS-ECF) evaluates visible, CDF, PIAFusion, and SPGFusion candidates with the target detector. A fused input is used only when its top-five confidence exceeds that of the visible image by a factor of 1.5; otherwise, the system preserves the visible input as a safety fallback.

Across LLVIP, M3FD, and an independent official-test subset of DroneVehicle, CS-ECF improved 9 of 12 detector checkpoints. The dataset-balanced AP50 gain was 0.0133, with a training-run-by-group hierarchical 95% interval of [0.0026, 0.0261]. Removing the shield increased the worst observed AP50 loss from 0.0040 to 0.0584. The two DroneVehicle checkpoints improved by 0.0251 and 0.0293; they belong to the same training run and are reported as correlated checks rather than independent architectures. A thermal-mask ablation further showed a small but consistent advantage over global detail injection.

Against the stronger Always CDF baseline, the dataset-balanced difference was 0.0102 AP50, with a training-run-by-group interval of [-0.00057, 0.02980]. The interval included zero; the point estimate was concentrated in M3FD and DroneVehicle and was not present on LLVIP. We therefore describe a risk-control and conditional-generalization contribution, not dominance over the best fixed fusion input.

Two COCO-pretrained evaluators that were not used in policy construction provided an independent check. CS-ECF improved AP50 over visible by 0.0213 and 0.0250, with scene-group intervals above zero; neither result separated from Always CDF. This supports portability of the visible-baseline gain while retaining the stated boundary.

We believe the work is relevant to the journal's readership because it addresses how thermal information should be admitted into a deployed detection system, uses real infrared–visible datasets, and reports both benefit and cost. The analysis also identifies an important boundary: M3FD gains are concentrated in the people class, and the LLVIP intervals cross zero. We therefore present CS-ECF as a risk-controlled evaluator-conditioned system, not as evidence of uniform improvement over all object classes.

The manuscript is original, has not been published previously, and is not under consideration by another journal. An earlier, related manuscript by the same authors on evaluator-dependent fidelity-index rankings was previously submitted to this journal and was not accepted. It is not under consideration elsewhere. The present manuscript is a distinct method study; it does not reproduce the earlier fidelity-ranking claims, and all shared datasets and evaluator infrastructure are identified in the manuscript. All authors have approved the submission and declare no competing interests. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors. Data and code availability are stated in the manuscript.

Thank you for your consideration.

Sincerely,

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